

Regression 2

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Topics for this lecture

- Significance of regression, the F-test
- Interaction effects
- Model checking
- Polynomial regression
- Transformations of the response
- Variable selection

The body data

Let's again look at the body data and the model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \epsilon_i$$

with the usual assumption of independent errors distributed as $\epsilon_i \sim N(0, \sigma^2)$. Here y_i is the body weight of person i , x_{1i} the height, x_{2i} is the circumference and x_{3i} is the age of the same person.

Our model with 3 predictors

```
##
## Call:
## lm(formula = Weight ~ Height + Circumference + Age, data = bodydata,
##     subset = 1:20)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.4088 -3.3900 -0.9123  3.0403  6.9827
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -66.5785    30.0098  -2.219  0.041330 *
## Height         0.4768     0.1600   2.981  0.008831 **
## Circumference  0.7769     0.1858   4.181  0.000706 ***
## Age          -0.2094     0.3731  -0.561  0.582352
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.259 on 16 degrees of freedom
## Multiple R-squared:  0.6852, Adjusted R-squared:  0.6262
## F-statistic: 11.61 on 3 and 16 DF,  p-value: 0.0002729
```

Prediction

A fitted model can be used for prediction of new observations

According to model 4, what is the predicted Weight of some person with these values?

```
##   Height Circumference Age
## 1    171           76  30
```

$$\$ = -66.579 + 0.4768 + 0.7769 - 0.2094 = 67.72 \$$$

```
newobs <- data.frame('Height'=171, 'Circumference'=76, 'Age'=30)
predict.lm(LinearModel.4, newdata=newobs)
```

```
##      1
## 67.7196
```

Other intervals in Prediction output

Confidence interval for $E(y|x_1, \dots, x_p)$

```
predict.lm(LinearModel.4, newdata=newobs,  
           level=0.95, interval=c("confidence"))
```

```
##           fit      lwr      upr  
## 1 67.7196 60.98252 74.45668
```

Prediction interval for $y|x_1, \dots, x_p$

```
predict.lm(LinearModel.4, newdata=newobs,  
           level=0.95, interval=c("prediction"))
```

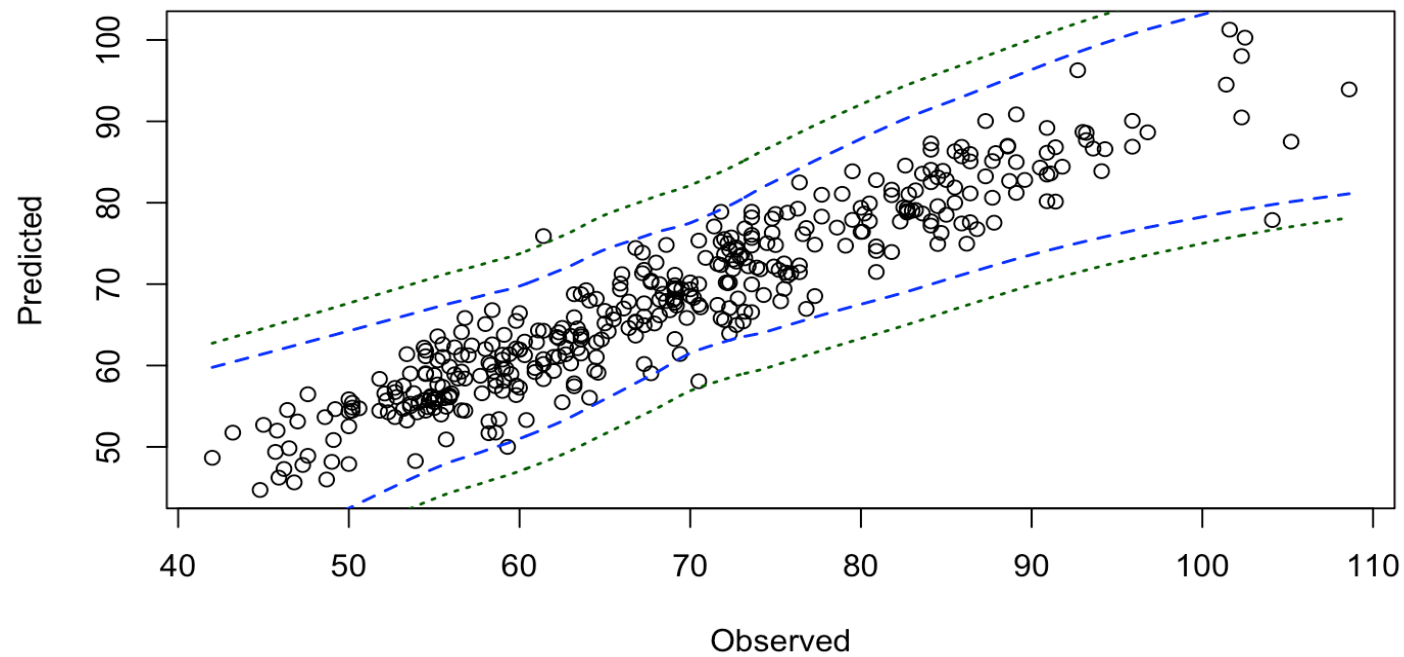
```
##           fit      lwr      upr  
## 1 67.7196 56.45463 78.98456
```

Compare real and predicted values

Let's predict the 387 extra observations and compare with actual observations!

##		fit	lwr	upr
##	25	73.94436	64.06824	83.82047
##	26	82.80148	72.72603	92.87693
##	27	81.03059	71.44393	90.61725
##	28	71.46852	61.81893	81.11811
##	29	65.93065	55.27452	76.58677
##	31	76.31124	66.90951	85.71296
##	32	69.35871	59.61763	79.09979
##	33	75.73802	65.67258	85.80346
##	34	82.52507	72.84459	92.20555
##	35	67.71321	57.86212	77.56431
##	37	73.86325	64.29154	83.43496
##	39	74.81258	65.46899	84.15617

Predicted vs observed - plot



Checking the prediction intervals

How many observations fell within the prediction intervals?

```
largerthanlower <- (newWeight > pred[,2])  
smallerthanupper <- (newWeight < pred[,3])  
inInterval <- which(largerthanlower & smallerthanupper)  
print(length(inInterval))
```

```
## [1] 381
```

Proportion of observations within prediction intervals:

```
length(inInterval)/length(newWeight)
```

```
## [1] 0.9844961
```

What should we expect?

Significance of regression?

We have seen how we can test the significance of single predictor terms with T-tests in the previous lesson.

We may also test the overall significance of the regression model.

Recall from earlier that $R^2 = \frac{SSR}{SST}$ is a measure of model fit. If the model can explain much of the observed variability in y , then SSR is close to SST and big compared to SSE.

Can be shown that a convenient test observator, typically named F (for Fisher) is:

$$F = \frac{SSR/p}{SSE/(n - p - 1)}$$

The F-test for overall significance of regression

Hypothesis:

$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$ vs $H_1 : \text{at least one } \beta_j \neq 0.$

Reject the null hypothesis at test level α if F is large, and more specifically if

$$F > F_{\alpha, p, n-p-1}$$

Let's see how this turns out for the body data model.

The F-test illustration

The body data

```
body1 <- lm(Weight ~ Height + Circumference + Age, data=bodydata)
Atab <- anova(body1)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: Weight
```

```
##           Df Sum Sq Mean Sq  F value    Pr(>F)
## Height      1  37360   37360 1960.906 < 2.2e-16 ***
## Circumference 1  26261   26261 1378.393 < 2.2e-16 ***
## Age         1    608     608   31.892 3.081e-08 ***
## Residuals   403   7678      19
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## SST = 71906.74 , SSR = 64228.7 ;
```

```
## Cumulative SST 37359.62 63621.07 64228.7 71906.74 ;
```

```
## F = 1123.73
```

The body data - Compare with previous

```
body1 <- lm(Weight ~ Height + Circumference + Age, data=bodydata)
Atab <- anova(body1)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: Weight
```

```
##           Df Sum Sq Mean Sq  F value    Pr(>F)
```

```
## Circumference    1   58175    58175 3053.453 < 2.2e-16 ***
```

```
## Height           1    5446     5446  285.846 < 2.2e-16 ***
```

```
## Age              1     608      608   31.892 3.081e-08 ***
```

```
## Residuals       403    7678         19
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## SST = 71906.74 , SSR = 64228.7 , F = 1123.73
```

Interaction effects

Interaction effects occur when the effect of one variable on the response variable depends on the level/value of another variable.

Consider the model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{2i} x_{3i} + \epsilon_i$$

here we have expanded the model with an interaction term between x_2 and x_3 (between circumference and age). How should we interpret this model?

How to fit the model in RStudio

```
data(bodydata)
body2 <- lm(Weight ~ Height + Circumference + Age +
            Circumference:Age, data=bodydata)
summary(body2)
```


The output

```
##
## Call:
## lm(formula = Weight ~ Height + Circumference + Age + Circumference:Age,
##     data = bodydata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1735  -2.5820  -0.1732   2.6033  22.5148
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -93.579875    6.030390  -15.518  < 2e-16 ***
## Height         0.414057    0.027243   15.199  < 2e-16 ***
## Circumference  1.244437    0.070990   17.530  < 2e-16 ***
## Age            0.611383    0.159085    3.843 0.000141 ***
## Circumference:Age -0.009476    0.001990   -4.763 2.67e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.252 on 402 degrees of freedom
## Multiple R-squared:  0.8989, Adjusted R-squared:  0.8979
## F-statistic: 893.8 on 4 and 402 DF,  p-value: < 2.2e-16
```

Model checking

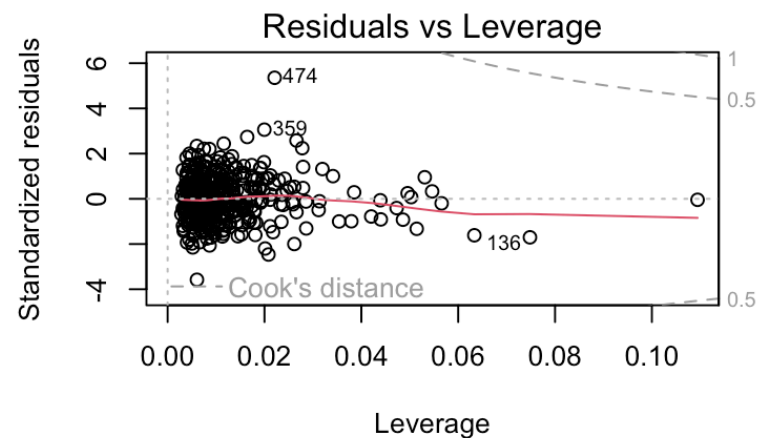
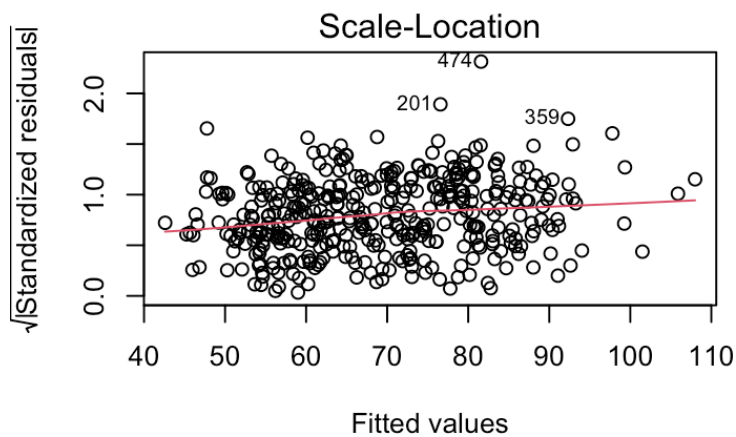
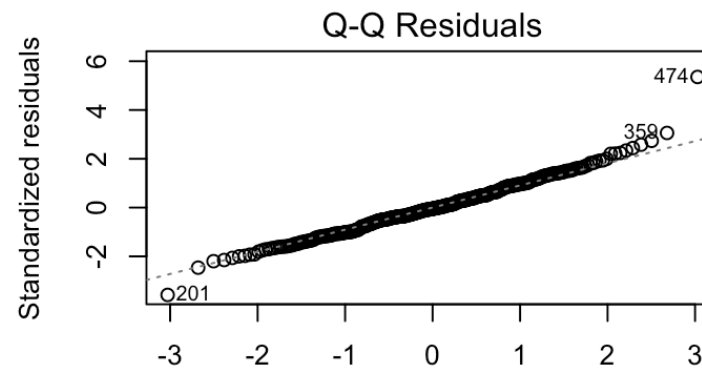
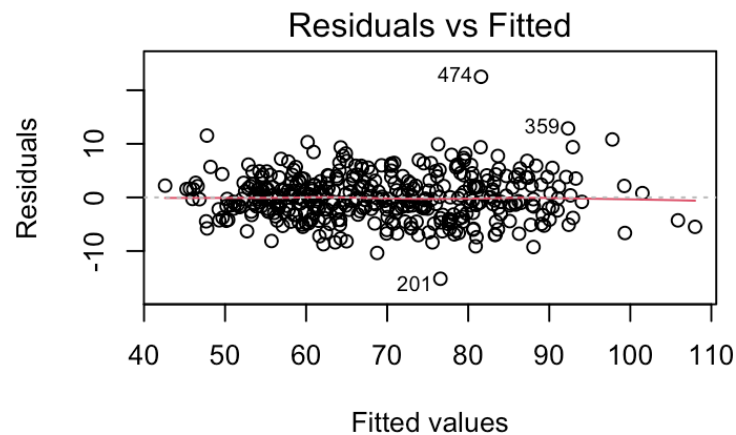
In standard linear regression we typically assume:

- Linearity
- Constant variance
- Independent observations
- Normality

We should check that these assumptions are OK. Typically we do this by examining the residuals

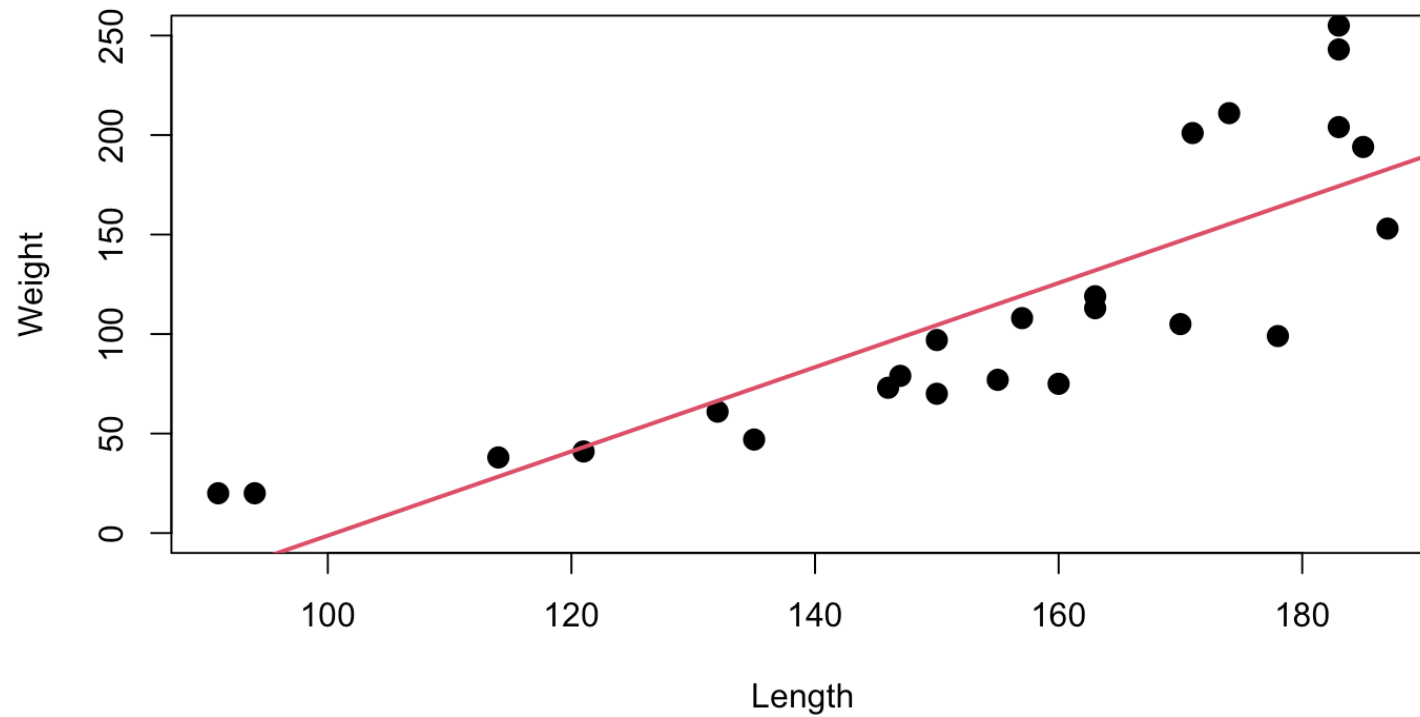
Residual plots

```
par(mfrow=c(2,2), mar=c(4.5,4,1.5,1.5))  
plot(body2, ask=FALSE)
```



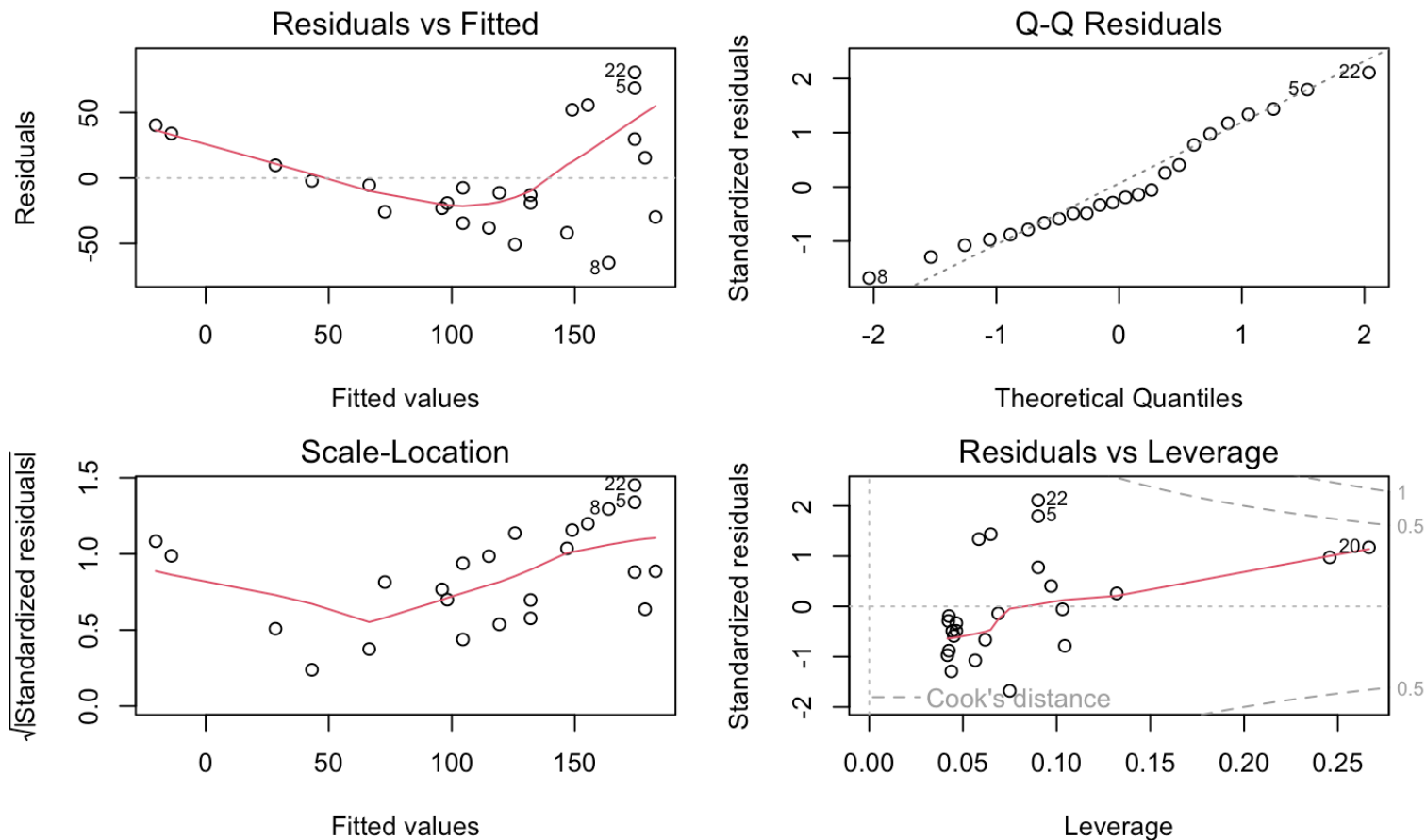
Explanations of the Q-Q plot

Another example - Bears



Residual plots

```
par(mfrow=c(2,2), mar=c(4.5,4,1.5,1.5))  
plot(bears1, ask=FALSE)
```



Outliers and influence statistics

- Leverage
- Standardized residuals
- Deleted residuals (PRESS residuals) (predict the deleted leave out one point)
- Cook's distance (combines leverage and standardized residuals) > 1 , can be dangerous

Explain leverage

Explain standardized residuals

Polynomial regression

Non-linear relationships may be modelled by polynomial regression.

Bear example

- y = Weight
- x_1 = Length
- x_2 = Length²

A multiple LINEAR model (linear in the β 's)

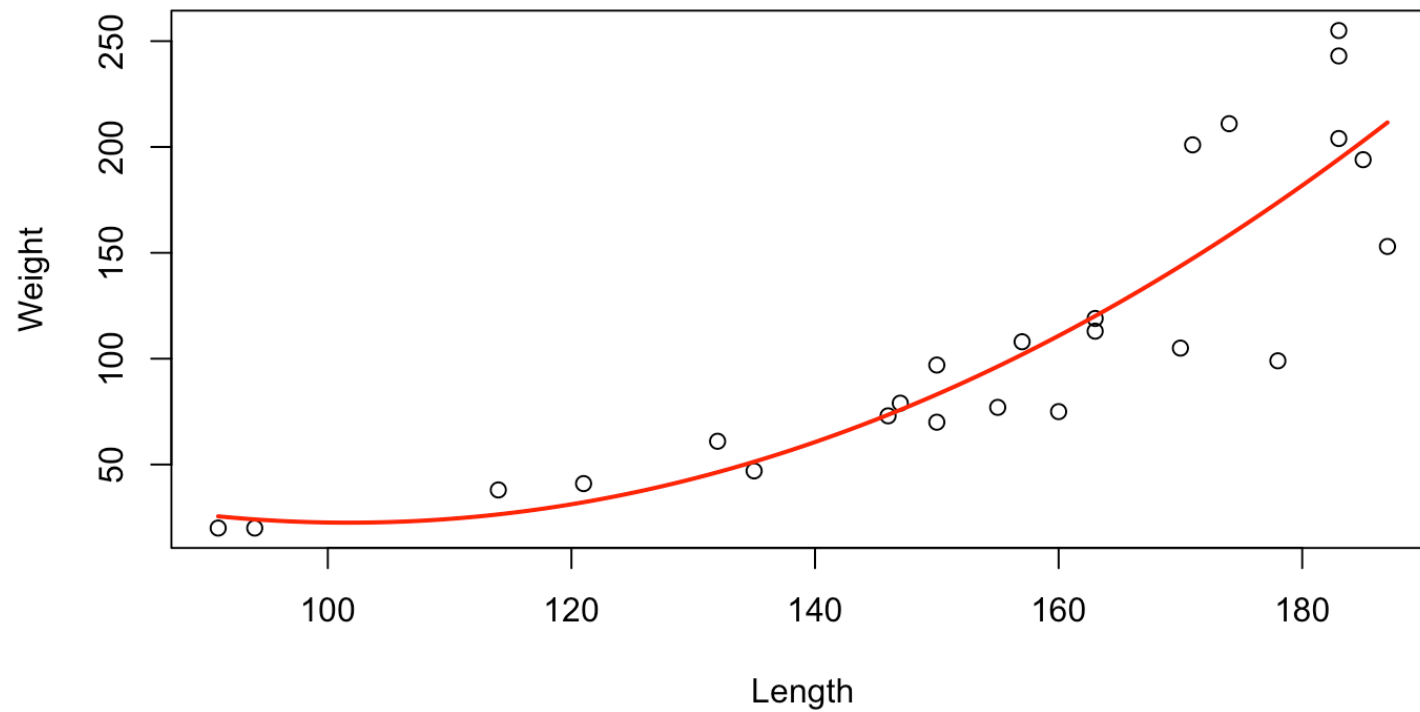
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

Fitting the model in R

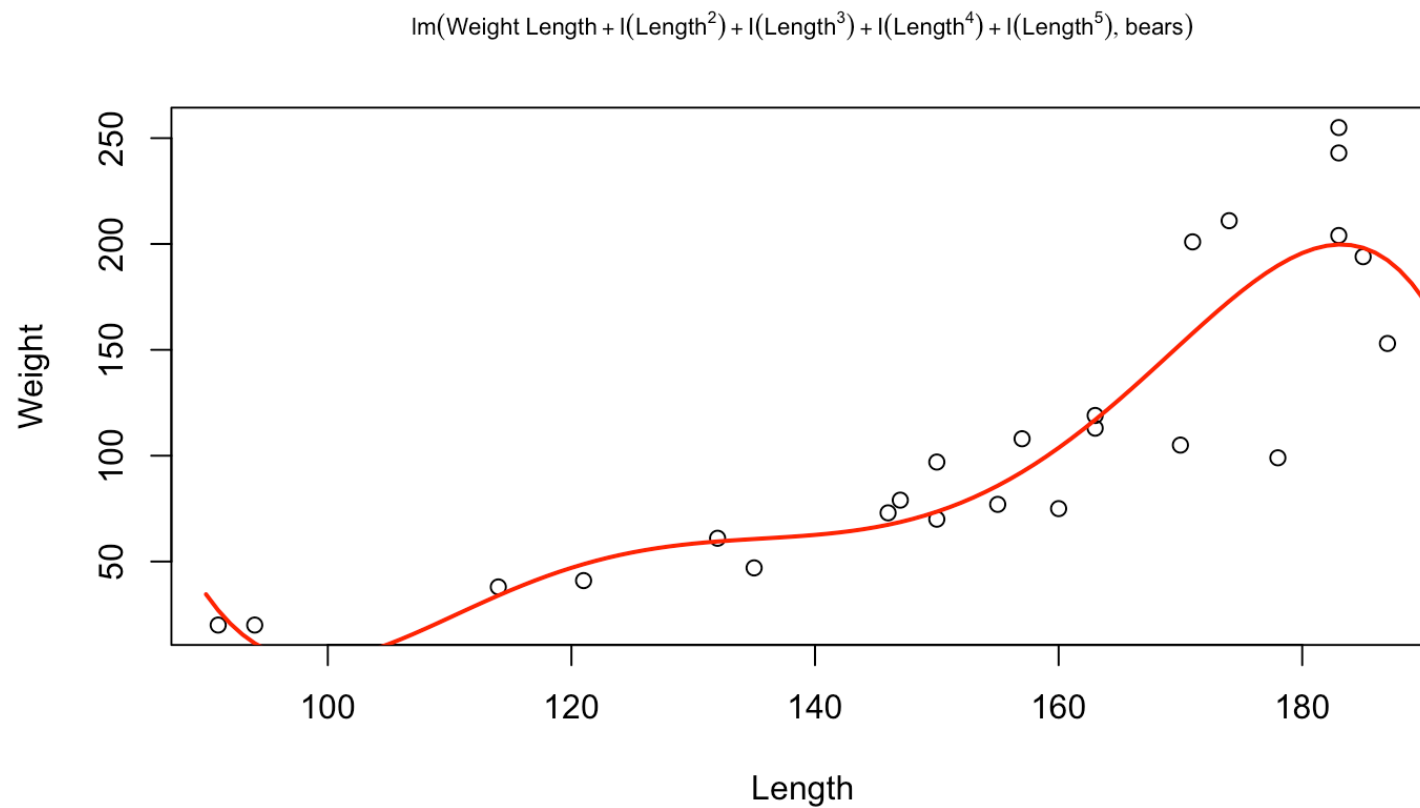
```
data(bears)
bears2 <- lm(Weight ~ Length + I(Length^2), data=bears)
summary(bears2)

##
## Call:
## lm(formula = Weight ~ Length + I(Length^2), data = bears)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -75.411 -10.514  -1.302   11.809   60.075
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  291.623760  175.892266   1.658  0.11219
## Length       -5.289645   2.525406  -2.095  0.04852 *
## I(Length^2)   0.026018   0.008825   2.948  0.00768 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 34.52 on 21 degrees of freedom
## Multiple R-squared:  0.7804 Adjusted R-squared:  0.7595
```

A fitted line plot

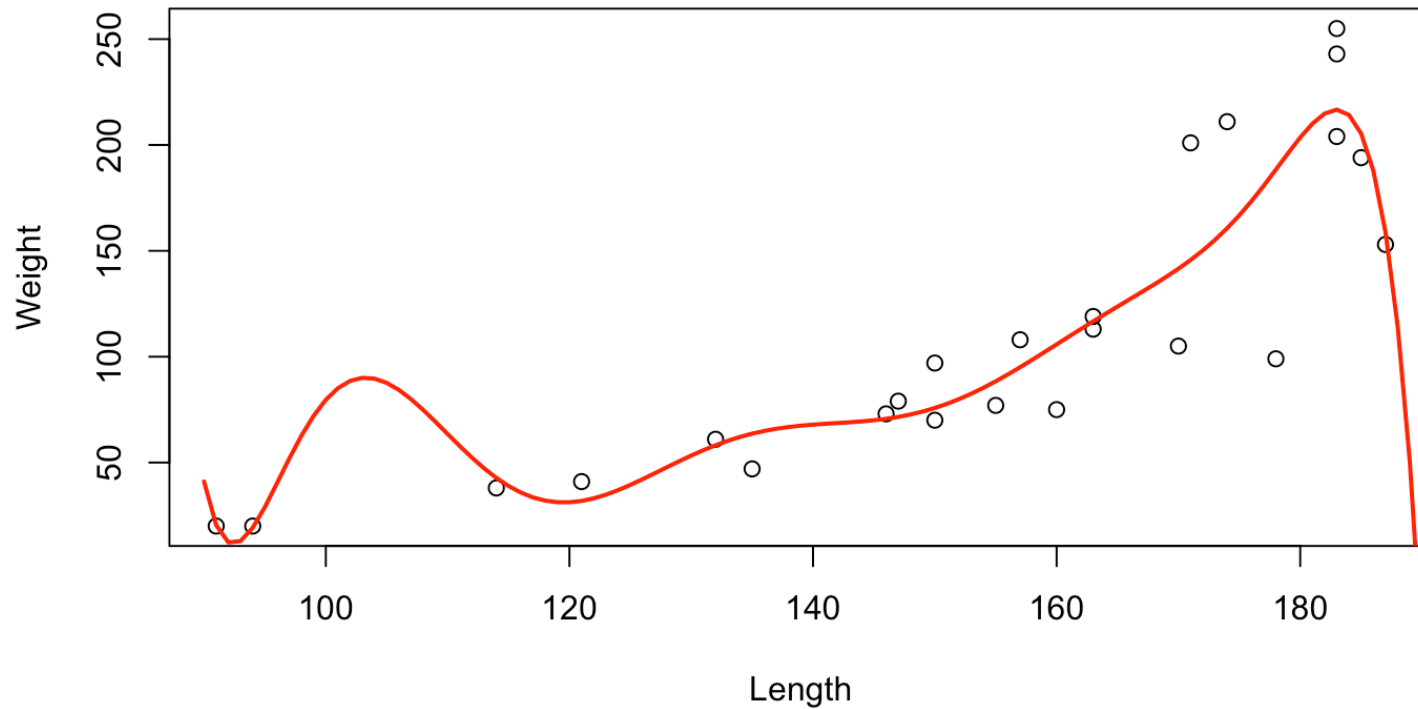


Careful! Over-fitting!

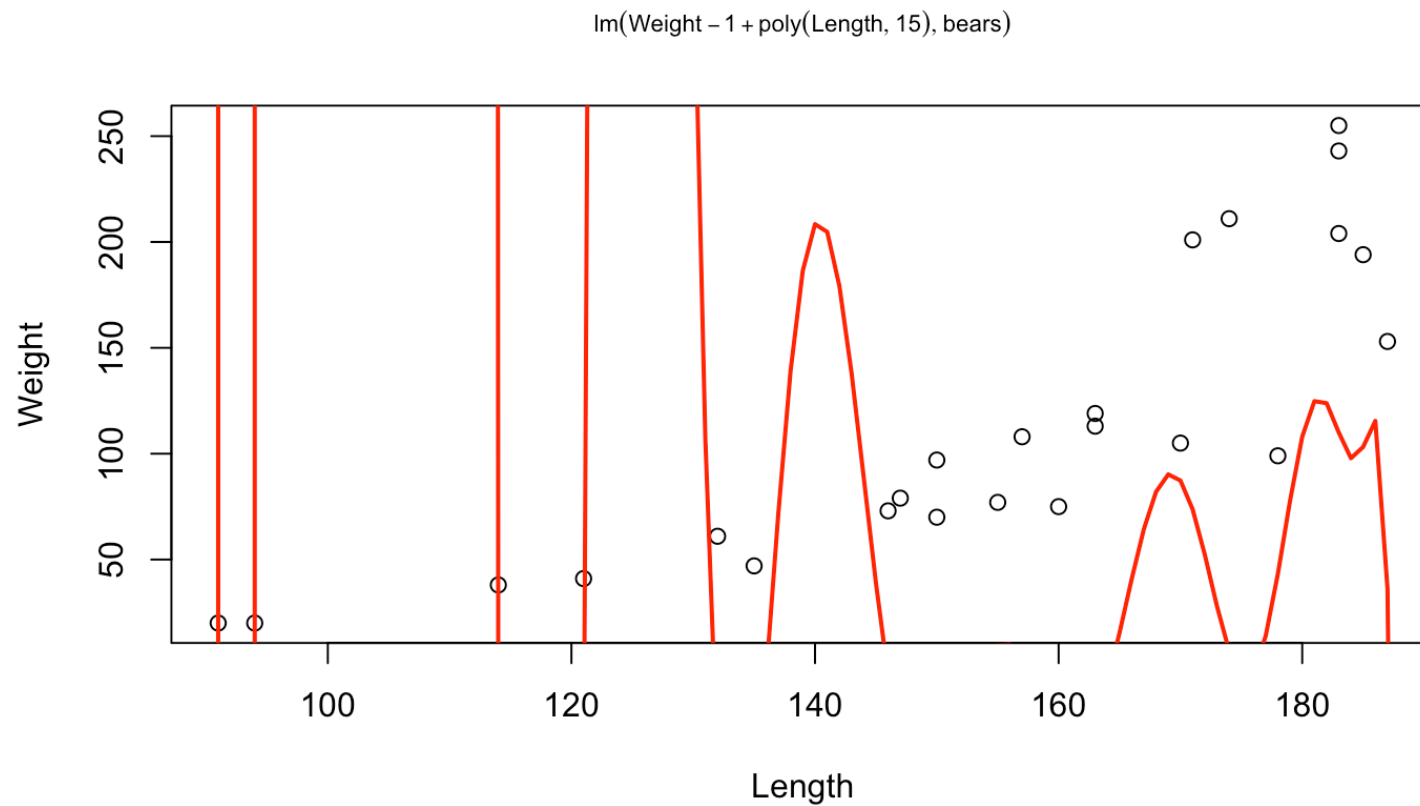


Careful! Over-fitting!

$\text{lm}(\text{Weight} \sim \text{Length} + \text{I}(\text{Length}^2) + \text{I}(\text{Length}^3) + \text{I}(\text{Length}^4) + \text{I}(\text{Length}^5) + \text{I}(\text{Length}^6) + \text{I}(\text{Length}^7) + \text{I}(\text{Length}^8) + \text{I}(\text{Length}^9) + \text{I}(\text{Length}^{10}), \text{bears})$



Careful! Over-fitting!



Transformations

Deviations from model assumptions may be fixed by transforming y and/or x .

Some transformations which may improve linearity and stabilize variance are

- $y' = \log(y)$
- $x' = \log(x)$
- $y' = \sqrt{y}$

Then use y' and/or x' instead of y and x to fit the model

The log(Weight) response model

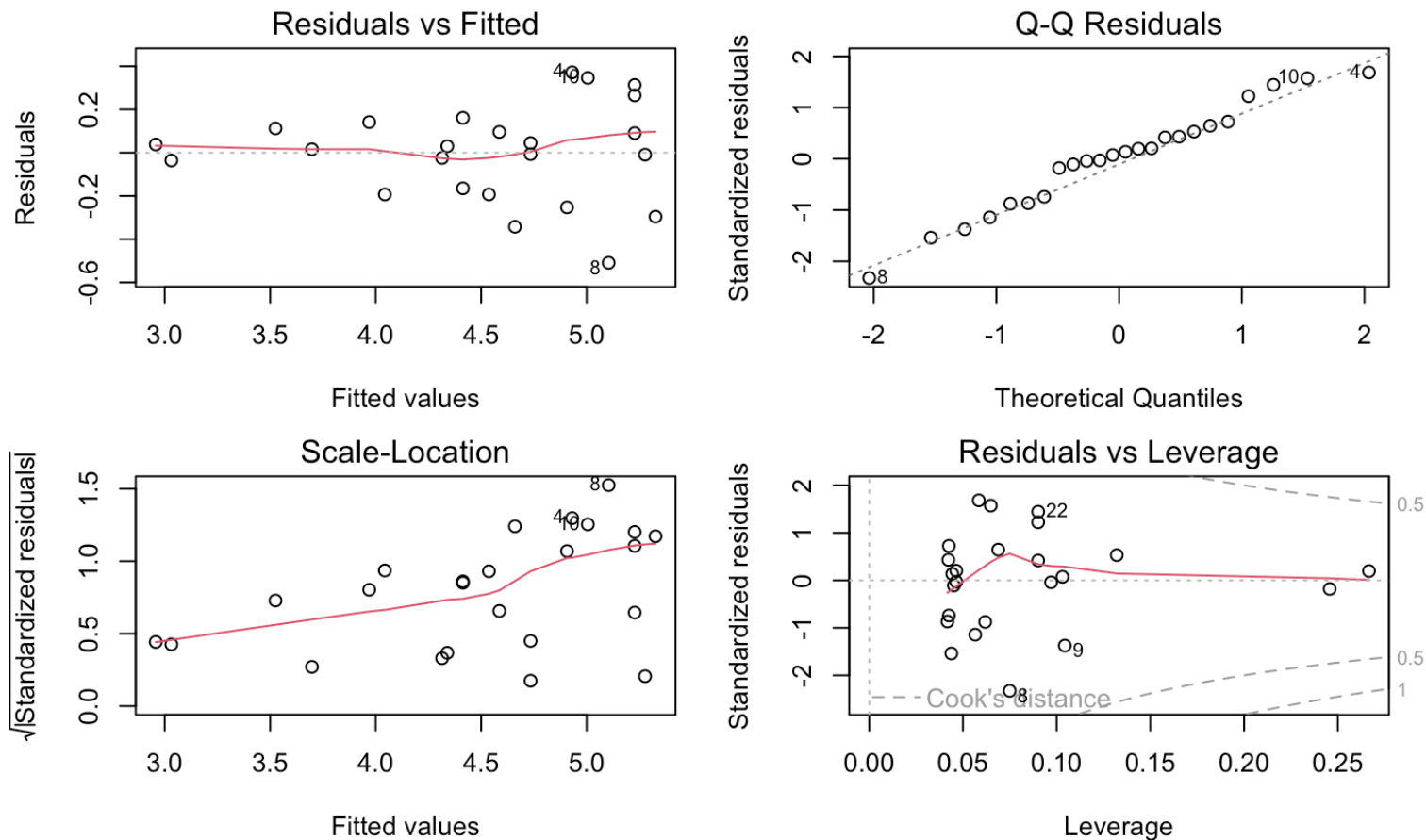
```
bears4 <- lm(log(Weight) ~ Length, data=bears)
summary(bears4)
```

```
##
## Call:
## lm(formula = log(Weight) ~ Length, data = bears)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.50918 -0.17192  0.02291  0.11978  0.37173
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.712120   0.268342   2.654   0.0145 *
## Length       0.024675   0.001718  14.363 1.17e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2276 on 22 degrees of freedom
## Multiple R-squared:  0.9036, Adjusted R-squared:  0.8992
## F-statistic: 206.3 on 1 and 22 DF,  p-value: 1.173e-12
```

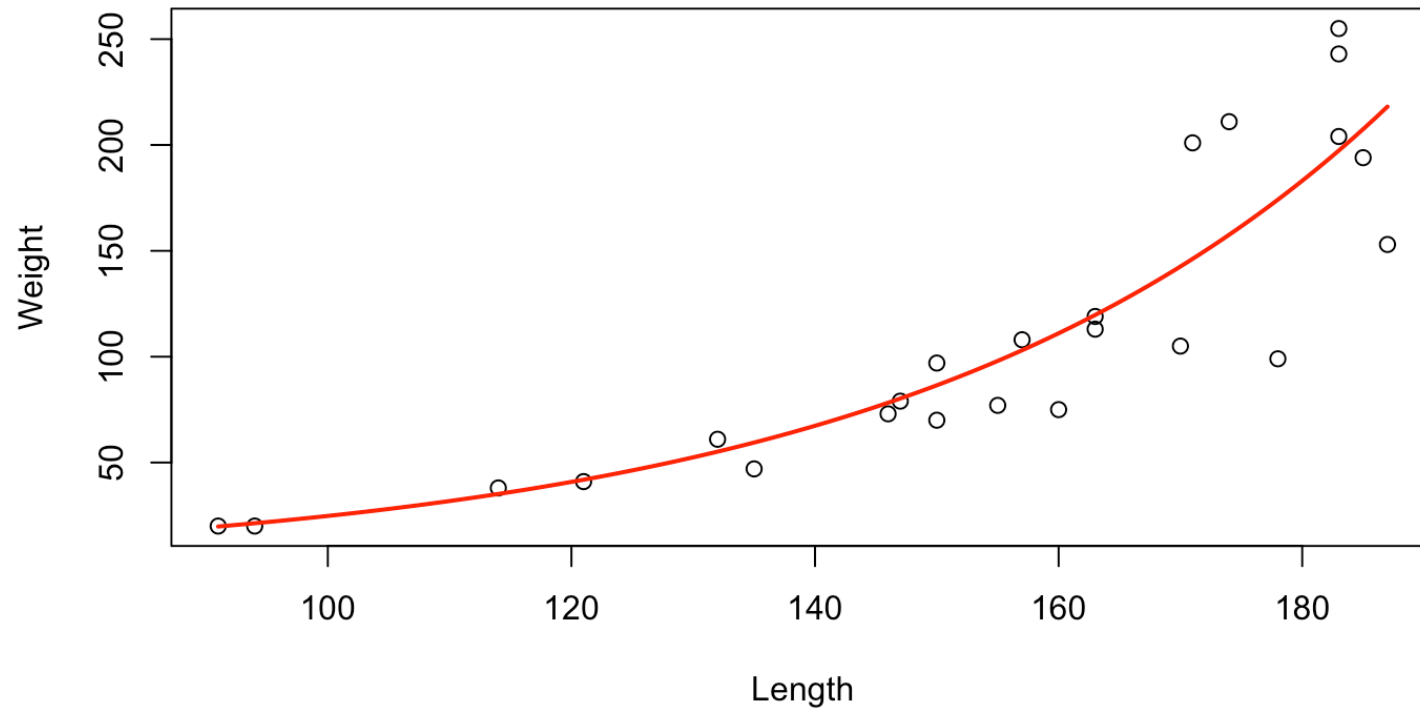
Explain the transformation and backtransformation

Residual plots

```
par(mfrow=c(2,2), mar=c(4.5,4,1.5,1.5))  
plot(bears4, ask=FALSE)
```



A fitted line plot



Variable selection

Variable selection is the process of selecting a subset of variables that are most relevant for the response in some way.

Various approaches. Some are based on hypothesis testing, some on some information criteria (like Akaike's Information Criteria, AIC)

Sometimes combined with stepwise approaches like:

- Forward selection
- Backward selection
- Forward/Backward
- Backward/Forward
- All subsets selection

Stepwise methods - Backward selection

```
data(bodydata)
library(mixlm)
body1 <- lm(Weight ~ Height*Circumference*Age, data=bodydata, subset=1:50)
backward(body1, alpha=0.05)
```

Stepwise methods - Forward selection

```
data(bodydata)
library(mixlm)
body1 <- lm(Weight ~ Height*Circumference*Age, data=bodydata, subset=1:50)
forward(body1, alpha=0.05)
```

Best subsets method

If the number of candidate variables, k , is limited, we may compare all possible models

There are 2^k possible models

Compare all model fits by R^2 , R_{adj}^2 and SSE (also called RSS)

Choose model based on:

- Goodness of fit
- Simplicity

Best subsets method - Example

```
data(bodydata)
library(mixlm)
body1 <- lm(Weight ~ Height*Circumference*Age, data=bodydata, subset=1:50)
best.subsets(body1, nbest=2)
```